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An Effective AODV-based Flooding Detection and Prevention for Smart Meter Network

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Abstract

In this paper, we investigate flooding attack detection and prevention for smart meter networks. The default Ad hoc On-Demand Distance Vector (AODV) protocol is vulnerable to flooding attacks because intermediate meters forward packets blindly. To detect and prevent flooding attacks, we propose a new AODV-based routing protocol called FLOoding aWareness AODV (FLOW-AODV) considering both with and without Internet Protocol (IP) spoofing. If no IP spoofing exists, we modify the route request forwarding procedure by adding new features. The main idea is to enable one-hop neighboring meters to detect attackers by counting the received route requests from the same source. Specifically, when a meter receives a request, it first determines whether it is a one-hop neighbor to the source or not. If yes, the one-hop neighboring meter updates its request counter for every received request from that source. When the received requests exceed a predefined number (e.g., 2), it perceives the source as an attacker and just discards the following requests, so the flooding is avoided. In this way, the receiver can prevent flooding attacks when there is no IP spoofing from flooding attacker. If IP spoofing exists, the flooding attacker will hide its real IP address. In this case, we have extended our algorithm and included trustworthiness of each neighbor in the neighbor routing table. Extensive simulations are conducted in NS3 to evaluate the performance of FLOW-AODV. Results show that FLOW-AODV obtains considerably higher performance in terms of packet delivery ratio, and average delay compared with the default AODV routing protocol. In addition, the FLOW-AODV significantly improves network performance and eliminates congestion when multiple attackers are involved.

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Keywords: FLOW-AODV; Flooding attack detection; Smart meter network; Packet delivery ratio;

1. Introduction

Smart grids, generally referred to as the next generation of power systems, rely on robust communication networks to provide efficient, secure, and reliable information delivery [1]. Smart meter networks play a critical role in smart grid communications, which are capable of communicating bidirectionally and efficiently between smart meters and

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utilities [2]. A typical smart meter network in a neighborhood consists of a large number of smart meters and several Data Aggregation Points (DAPs). DAPs are responsible for collecting data from their associated smart meters and then assist in delivering data to utilities.

Smart meter networks commonly form a mesh mode to provide self-adapting and multi-hop communications as well as to compensate for failures in case of a break-down of relay meters [3]. Due to the inherent mesh topology, a routing protocol is required for smart meter networks to find a path from a smart meter to an associated DAP. In recent years, Ad hoc On-Demand Distance Vector (AODV) routing protocol is recommended because it meets the requirements of on-demand operation in a smart meter networks. However, the AODV is vulnerable to different kinds of Denial-of-Service (DoS) attacks including blackhole, gray hole, and flooding. Under attacks, packets are frequently dropped and the data transmission can be delayed, corrupted or even blocked, which significantly damages the network performance. Therefore, improving the security of smart meter networks with a mesh topology is a critical issue.

In literature, several AODV-based algorithms are proposed for the detection of Route Request (RREQ) flooding attack. The majority of these algorithms assign a threshold or predefined RREQ rate limit (i.e., a maximum number of RREQ can be initiated from a node during a certain time interval) to control the flow of RREQ message [4]. However, if flooding attacker already learns the RREQ rate limit, it can continue flooding by controlling flooding frequency. Therefore, the algorithms based on RREQ rate limit are not effective to prevent flooding in smart meter networks. Moreover, the existing algorithms do not take into account the ability of attackers to control the flooding frequency. In this paper, we address this challenge by assuming that the attackers can control the flooding frequency intelligently, and then generate an appropriate number of RREQ like a normal meter within a certain interval. We propose a new AODV-based routing protocol called FLOoding aWareness AODV (FLOW-AODV) to detect and prevent flooding attacks in smart meter networks. The FLOW-AODV improves the RREQ forwarding procedure of AODV by adding new functions, so it is able to preserve the normal flow of traffic regardless of flooding frequency or the number of attackers throughout the network. The main contributions of this paper are highlighted as follows.

The first contribution is identifying one-hop neighbors from a source meter by observing RREQ messages. Note that intermediate meters are forwarding meters. A forwarding meter compares two IP Addresses from its immediate sender (i.e., from which the RREQ comes in) and the originator (i.e., actual source), which are contained in the RREQ message. If these two are the same, a forwarding meter identifies itself as one-hop neighbor. Note that every smart meter maintains an RREQ counter entry in its routing table, but only a one-hop neighbor will update its counter entry for every fresh RREQ. The fresh RREQ is determined by monitoring the originator's sequence number in the RREQ message.

The second contribution is proposing a novel approach to detect and prevent flooding attacks in the first place. To be specific, one-hop neighbors check the counter entry for received RREQs in its routing table. The source meter will be marked as a malicious meter if the number of RREQs exceeds a threshold and the following RREQs are rejected.

The third contribution is preventing flooding attack with IP spoofing by maintaining a trustworthiness value of each neighbor in the neighbor routing table. Before forwarding any RREQ, each meter updates the trustworthiness entry and checks whether it is below a threshold or not. If it is assertive, the receiver meter will avoid the following RREQs of immediate sender.

The fourth contribution is to evaluate the performance of FLOW-AODV compared with the default AODV based on a real suburb neighborhood topology. Simulation results verify that FLOW-AODV significantly improves the network performance in terms of packet delivery ratio (PDR) and average delay.

The rest of this paper is organized as follows. In Section 2, related works on flooding attack detection and some unsolved issues of those works are introduced. Section 3 describes the proposed FLOW-AODV. Section 4 presents the simulation parameters and performance metrics. The performance of FLOW-AODV is evaluated and compared with the default AODV in Section 5. Section 6 draws the conclusions.

2. Related Works

The detection and prevention of flooding attack in smart grid are under explored and only limited work can be found in the literature. In [3], authors propose some distributed intrusion detection modules using support vector machine and artificial immune system which can detect and classify malicious data and possible attacks. However, the synchronization among different modules enhances the system complexity. In [5], flooding attack is implemented on the utility server in smart meter network. An attacker is connected to a bot network (botnet) which consists of

bots (i.e., devices which are used by an attacker for malicious activities). This work only considers that the server is flooded by User Datagram Protocol (UDP) packets from an attacker. In [6, 7], authors introduce a newly discovered denial of service attack, called puppet attack in the smart grid network. The attacker sends puppets data packets with error routes to nodes that generates a high volume of route packets.

AODV has been widely used in different networks and the flooding attack has received considerable attention. In [8, 9], a filtering technique is adopted in mobile ad hoc networks running AODV routing protocol. In this technique, a threshold is assigned in order to limit the RREQ packets for a certain period of time. In [10], a neighbor suppression algorithm is developed to prevent the flooding attack. In such an algorithm, the neighbor node assigns a priority for all of its immediate sources and based on the priority it forwards a particular RREQ. Every node maintains a threshold to block the flooding RREQ. Moreover, if a malicious node knows the threshold, it can disguise itself a high priority node of neighbors by adjusting the flooding frequency. In [11], an algorithm is developed to detect and isolate the flooding attacker based on three parameters: RREQ Rate Limit, RREQ Blacklist Limit, and RREQ Accept Limit. If all nodes check the parameters before forwarding the RREQ packet, the end-to-end delay of a network will increase significantly. In [12], flooding attack prevention scheme is proposed by introducing a new routing table, called Avoiding Mistaken Transmission Table (AMIT). However, maintaining this routing table consumes more communication resources and the checking of AMIT table slows down the communication process. In [13, 14], the authors introduced a trust-based scheme for RREQ Flooding Attack in mobile ad hoc network. Based on the trust level, each node finds out the relationship with neighbors. This scheme assigns a separate threshold for different neighbor nodes. However, if a malicious node is a customized normal node or it somehow manages the authenticity, the neighbor node will place a high trust level for malicious nodes that results in forwards the flooding RREQs.

In our work, we extensively examine the flooding attacks from compromised meters under ADOV in smart meter network. Specifically, we focus on RREQ flooding attacks.

3. FLOW-AODV

In this section, we will introduce our proposed approach, termed FLOW-AODV, to protect a smart meter network from flooding attack. As we have mentioned previously, in the default AODV, forwarding meters just check the source sequence number of RREQ, which provides opportunities for flooding attack. We propose to modify the RREQ forwarding system of the default AODV so as to effectively prevent flooding attacks. The details are presented as follows based on two different scenarios:

3.1. FLOW-AODV without IP Spoofing from Flooding Attacker

In this subsection, we first consider the scenario without IP spoofing. That is, we assume the flooding attackers generate fake RREQs without changing the IP address field of the sender in the packet header. Based on this assumption, the detection and prevention procedures of FLOW-AODV are described below:

First, every meter maintains an RREQ counter entry in its routing table. When a one-hop neighbor meter receives an RREQ from the source meter, the one-hop neighbor increases the RREQ counter. Note that only one-hop neighbor will count the RREQ number before forwarding any RREQ while the other forwarding meters just check the sequence number to distinguish between an updated and duplicate RREQ. Any forwarding meter determines whether it is a one-hop neighbor to the source by comparing the originator IP address and the IP address of the immediate sender. If equal, the forwarding meter will consider itself as a one-hop neighbor to the source and then increases the RREQ counter when receiving more RREQ from the same source. In FLOW-AODV, only one-hop neighboring meters need to count RREQs, so there is no extra overhead generating for other meters.

Second, when a one-hop neighbor meter receives an RREQ, it checks the RREQ counter entry and source sequence number. As we know, flooding attacker can send fake RREQs with an updated source sequence number. Therefore, checking RREQ counter is the only way to detect and avoid flooding attackers. Specifically, when a one-hop neighbor meter receives more than a predefined number of RREQs from the same source, it just discards the following RREQs and perceives the source as an attacker. In our paper, we set the maximum two RREQs for a particular destination because in the default AODV, a source meter can initiate up to two RREQs for the same destination and update the sequence number of RREQ only for retransmission. This implies that for a normal meter, if the first RREQ is lost, it will resend only once. Therefore, if a source meter sends more than two RREQs, it can be viewed as malicious

flooding attacker. In this way, one-hop neighbor meters are able to detect a malicious meter and stop forwarding more fake RREQs. At the same time, the malicious IP address will be recorded in blacklist routing table. Moreover, if the malicious meter changes its position and IP address, the surrounding one-hop neighbor will effectively detect it using the same way and ignores RREQs from it.

To facilitate better understanding, Fig. 1 describes a flooding attack detection where SM denotes a smart meter. SM1 to SM8 are one-hop neighbors to the malicious meter and they increase the counter for received RREQ every time from the malicious meter. In contrast, SM9 to SM16 are the normal forwarding meters and they forward any fresh RREQ without changing the RREQ counter. SM1-SM8 are all able to identify the malicious meter after receiving the third RREQ. It can be seen that even though the malicious meter can generate multiple fake RREQs, one-hop neighbor only accepts the first two RREQs and stop forwarding the following RREQs to their neighbors (i.e., SM 9-16). As a result, the whole network will not be affected by the flooding attack after receiving two fake RREQs.

3.2. FLOW-AODV with IP Spoofing from Flooding Attacker

In this subsection, we assume the attacker can alter the sender IP address, which is referred to as IP spoofing in RREQ packet header and hide its own IP address. Because of IP spoofing, one-hop neighbor cannot prevent RREQ flooding using IP address to count the received RREQs. Therefore, we extend our algorithm by tracking RREQ rate of an immediate sender under each smart meter. Besides that, we have assigned three different ranges of the RREQ rate which are high rate (low flooding interval), intermediate rate and low rate (high flooding interval) as shown in Fig. 2. Based on the RREQ rate, each smart meter evaluates the trustworthiness of its neighbor. The details are presented below:

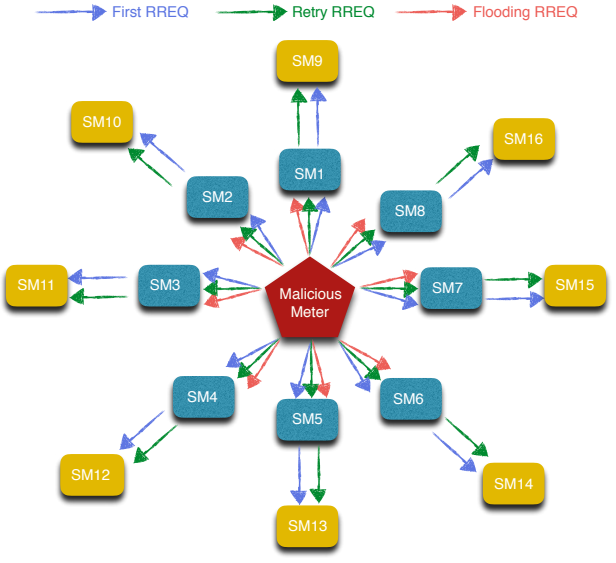


Figure 1: Illustration of flooding detection in FLOW-AODV without IP spoofing from attacker.

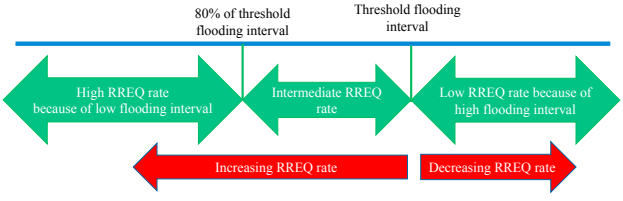


Figure 2: Illustration of the range of RREQ rate for preventing IP spoofing flooding attack.

Step 1: Update the number of connected one-hop neighbor for every Hello Packet communication.

Step 2: Every meter maintains a trustworthiness value of one-hop neighbor in its neighbor routing table. The trustworthiness value of one-hop neighbor is assigned based on RREQ rate.

Initially, the total probability of trustworthiness of any one-hop neighbor (denoted by P_{Total}) and the probability of trustworthiness is 1. As time progresses, if the current RREQ rate of any neighbor is in the high rate region, then it updates the trustworthiness as follows:

Probability of loss of trustworthiness,

$$P_{LT} = \frac{1}{Number\ of\ connected\ one\text{-}hop\ neighbor}$$

Probability of trustworthiness,

$$P_T = P_{Total} - P_{LT}.$$

Because of the loss of trustworthiness, P_{Total} is updated from the current trustworthiness (P_T) and included as an updated trustworthiness in the neighbor routing table. Therefore, this updating process will continue until the node trustworthiness belongs to a certain limit. However, if the trustworthiness (P_T) of any immediate sender is below 20% in the receiver neighbor table, then the receiver will not further forward any RREQ from that sender.

Step 3: When the immediate sender generates RREQ in the intermediate rate, the receiving meter first calculates the deviation from threshold rate and counts how many times its rate falling in the intermediate rate region. Deviation of RREQ rate is defined as:

$$D_R = \frac{Threshold\ rate, Th_R - Present\ rate, P_R}{Threshold\ rate, Th_R}.$$

Note that, the value of D_R is varied from 0 to 0.2. For a faster flooding prevention, we have considered the maximum value of D_R. The trustworthiness of immediate sender is updated as:

$$P_T = 1 - max(D_R) * N.$$

Here, N is the number of times the sender RREQ rate in the intermediate region. Whenever, the receiver meter shows that P_T is below 20% for an immediate sender, it prevents the flow of further any RREQ through that link. Note that under smart meter network scenario, the threshold rate cannot degrade the performance and should sustain almost the normal traffic throughout the network. Therefore, in our simulations, we will not consider any preventions for low RREQ flooding.

4. Simulation Parameters and Performance Metrics

To evaluate the performance of FLOW-AODV, we conduct extensive simulations on Network Simulator 3 (NS3). In the simulation, an actual suburban neighborhood with 100 houses is selected from Rapid City, South Dakota, USA. It is assumed that each house installs a smart meter and a DAP is placed at the center of the network. The longitude and latitude of each house are obtained from Google map and the distance between any two houses is calculated by the Haversine formula [15]:

$$\begin{aligned} a &= \sin^2\left(\frac{\Delta lat}{2}\right) + \cos(lat1)\cos(lat2)\sin^2\left(\frac{\Delta lon}{2}\right) \\ c &= 2\arctan 2(\sqrt{a}, \sqrt{1-a}) \\ d &= R \times c \\ \Delta lat &= |lat2 - lat1| \\ \Delta lon &= |lon2 - lon1| \end{aligned} \tag{1}$$

where $R = 6373$ km, is the radius of Earth. The transmission power of a meter, which is denoted by P can be calculated through the following equation [16][17]:

$$P = \left(\frac{4\pi D}{0.12476}\right)^2 \times 10^{(-12.5)} \tag{2}$$

where $D = 55$ meters, is the required transmission range of a smart meter.

Generally, three metrics are used to assess the performance of FLOW-AODV, which are the PDR, average throughput and average end-to-end delay [18]. However, in our simulation, we only presented PDR and average delay metrics.

a. *PDR*: It is the ratio of the number of packets received by DAP to that of sent by a smart meter. For the entire network, the overall PDR is defined as the total number of packets received by DAP to that of sent by all smart meters.

b. *Average throughput*: It is the ratio of total data (in bits) received by DAP to overall time consumed on all successful transmissions.

c. *Average end-to-end delay*: It is the average time taken by a data packet transmitted from a smart meter to DAP. It is composed of the delay caused by the process of route discovery and the queue delay before a data transmission. In Table 1, we list all settings applied in our simulations.

5. Simulation Results

In this section, we evaluate the performance of FLOW-AODV and default AODV based on two important performance metrics (i.e., overall PDR and average end-to-end delay). Both performance metrics are assessed by changing the flooding interval and number of attacker.

5.1. Performance Comparison between AODV and FLOW-AODV by Changing RREQ Flooding Interval

First, we consider a single flooding attacker that locates close to the DAP, and smart meters in the network communicate with the DAP one by one. The performance of the FLOW-AODV with different flooding intervals is compared with the default AODV protocol.

The impact of flooding interval on PDR of the default AODV and the FLOW-AODV protocols are presented in Fig. 3. The successful transmission rate of a real RREQ reduces significantly due to the high collision probability caused by fake RREQ messages. Moreover, due to the traffic congestion, a smart meter needs to spend more time on finding an appropriate routing path, which increases the end-to-end delay of a data transmission considerably.

Table 1: Specific parameter setup for simulations

Items	Attributes
Simulator	NS3
Number of Data Concentrator	1
Transmission Range	55m
Routing Protocol	AODV and FLOW-AODV
Number of Meters	100
Number of Malicious Meter	1 to 5
Mobility Model	Static
Traffic	Constant Bit Rate
MAC/PHY	IEEE 802.11b
Number of Sending Packet by Each Meter	1
Packet Interval	10 Seconds
Data Packet Rate	11Mbps
Control Packet Rate	1Mbps
Packet Size	1KB

According to the simulation results, an attacker harms the performance of a smart meter network more severely when the flooding interval decreases from 20 ms to 2 ms. To be specific, as depicted in Fig. 3, when the flooding interval of an attacker is 2 ms, the PDR under the default AODV is only 2.02%. By contrast, the FLOW-AODV is almost immune to the RREQ flooding attack even when the flooding interval is very short. However, under IP spoofing, the performance of FLOW-AODV slightly degrades because of increasing control packet overhead.

Similar to Fig. 3, the end-to-end delay of the default AODV is vulnerable to a high frequent flooding attack as shown in Fig. 4. For example, more than 1000 ms average delay is caused when the flooding interval is less than 20 ms.

In other words, a flooding attacker is able to interrupt the communications severely between smart meters and DAP under higher flooding frequency. In comparison, the FLOW-AODV can reduce the end-to-end delay of data transmission dramatically even if under an attack with a short flooding interval.

5.2. Performance Comparison between AODV and FLOW-AODV by Changing Number of Attackers

Next, we fix the flooding interval to 10 ms, and evaluate how the number of attackers affects the PDR and average delay of the FLOW-AODV and the default AODV protocols.

In Fig. 5, multiple attackers are placed randomly in a network to compare the performance between the default AODV and the FLOW-AODV protocols. As illustrated in the figure, with the increased number of attackers, the performance of the default AODV declines from all aspects. For example, when the number of attackers increases from 1 to 5, the PDR of the default AODV decreases from 80% to 1%. By contrast, in the FLOW-AODV, the PDR almost reaches 100%.

Fig. 6 depicts the average delay comparison between default AODV and FLOW-AODV under multiple flooding attackers. In default AODV, the overall average delay is more than 1500ms that exceeds the upper limit of tolerance in smart meter network. Because of heavy traffic congestion, a sender has to wait for a longer time to find a routing path. If any sender fails to find a routing path, it just drops the incoming packets after timeout. By contrast, the FLOW-AODV can mitigate the congestion problem efficiently. As demonstrated in the figure, the average delay in the FLOW-AODV is always less than 200 ms no matter how many attackers are placed in the network, since the FLOW-AODV can filter out the redundant fake RREQs transmitted by attackers in the first place.

6. Conclusions

In this paper, we have introduced a promising approach FLOW-AODV to detect and prevent RREQ flooding attack in smart meter network. By considering an actual suburban neighborhood topology, the performance of FLOW-AODV and default AODV is comprehensively evaluated based on PDR and average end to end delay. All performance parameters are examined by changing the flooding interval and number of attacker.

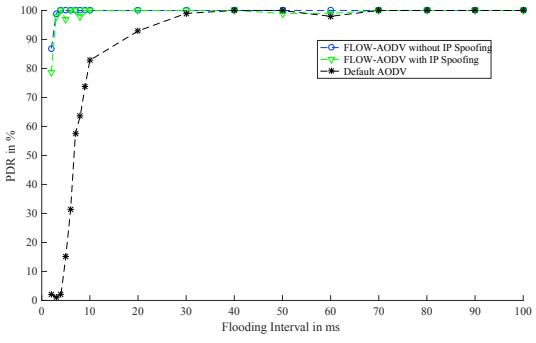


Figure 3: Effect of PDR with flooding interval.

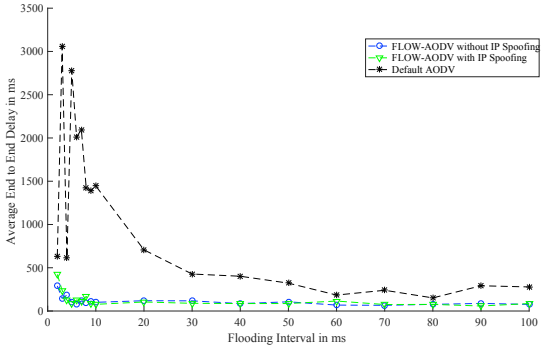


Figure 4: Effect of average delay with flooding interval.

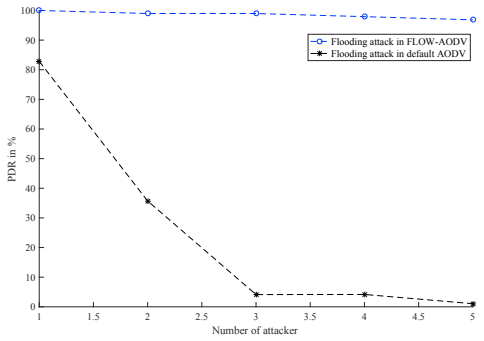


Figure 5: Effect of multiple flooding attackers on PDR.

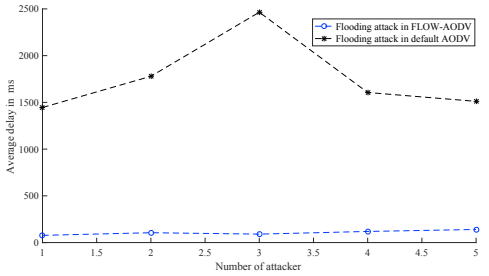


Figure 6: Effect of multiple flooding attackers on average delay.

Simulation results reveal that for smaller flooding interval or higher flooding frequency, FLOW-AODV significantly improves the performance metrics compared with default AODV. Specifically, smaller average delay in FLOW-AODV implies that it prevents the redundant RREQ flooding under frequent flooding attack. Furthermore, simulation results based on the appearance of multiple flooding attackers show that FLOW-AODV maintains almost 100% PDR and less than 200ms delay. That indicates FLOW-AODV is able to prevent flooding attacks effectively regardless of the number of attackers.

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